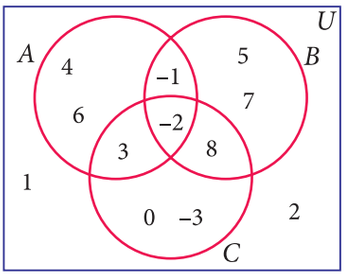
## Exercise 1.5

1. Using the adjacent Venn diagram, find the following sets:




(i) $A - B$ (ii) $B - C$ (iii) $A' \cup B'$

(iv) $A' \cap B'$ (v) $(B \cup C)'$

(vi) $A - (B \cup C)$ (vii) $A - (B \cap C)$

2. If $K = \{a, b, d, e, f\}$, $L = \{b, c, d, g\}$ and $M = \{a, b, c, d, h\}$, then find the following:

(i) $K \cup (L \cap M)$ (ii) $K \cap (L \cup M)$

(iii) $(K \cup L) \cap (K \cup M)$ (iv) $(K \cap L) \cup (K \cap M)$

and verify distributive laws.

3. If $A = \{x : x \in \mathbb{Z}, -2 < x \leq 4\}$, $B = \{x : x \in \mathbb{W}, x \leq 5\}$, $C = \{-4, -1, 0, 2, 3, 4\}$, then verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

4. Verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ using Venn diagrams.

5. If $A = \{b, c, e, g, h\}$, $B = \{a, c, d, g, i\}$ and $C = \{a, d, e, g, h\}$, then show that $A - (B \cap C) = (A - B) \cup (A - C)$.

6. If $A = \{x : x = 6n, n \in \mathbb{W} \text{ and } n < 6\}$, $B = \{x : x = 2n, n \in \mathbb{N} \text{ and } 2 < n \leq 9\}$ and $C = \{x : x = 3n, n \in \mathbb{N} \text{ and } 4 \leq n < 10\}$, then show that $A - (B \cap C) = (A - B) \cup (A - C)$

7. If $A = \{-2, 0, 1, 3, 5\}$, $B = \{-1, 0, 2, 5, 6\}$ and $C = \{-1, 2, 5, 6, 7\}$, then show that $A - (B \cup C) = (A - B) \cap (A - C)$.

8. If $A = \{y : y = \frac{a+1}{2}, a \in \mathbb{W} \text{ and } a \leq 5\}$, $B = \{y : y = \frac{2n-1}{2}, n \in \mathbb{W} \text{ and } n < 5\}$ and $C = \left\{-1, -\frac{1}{2}, 1, \frac{3}{2}, 2\right\}$, then show that $A - (B \cup C) = (A - B) \cap (A - C)$.

9. Verify $A - (B \cap C) = (A - B) \cup (A - C)$ using Venn diagrams.

10. If $U = \{4, 7, 8, 10, 11, 12, 15, 16\}$, $A = \{7, 8, 11, 12\}$ and $B = \{4, 8, 12, 15\}$ then verify De Morgan's Laws for complementation.

11. Verify $(A \cap B)' = A' \cup B'$ using Venn diagrams.



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